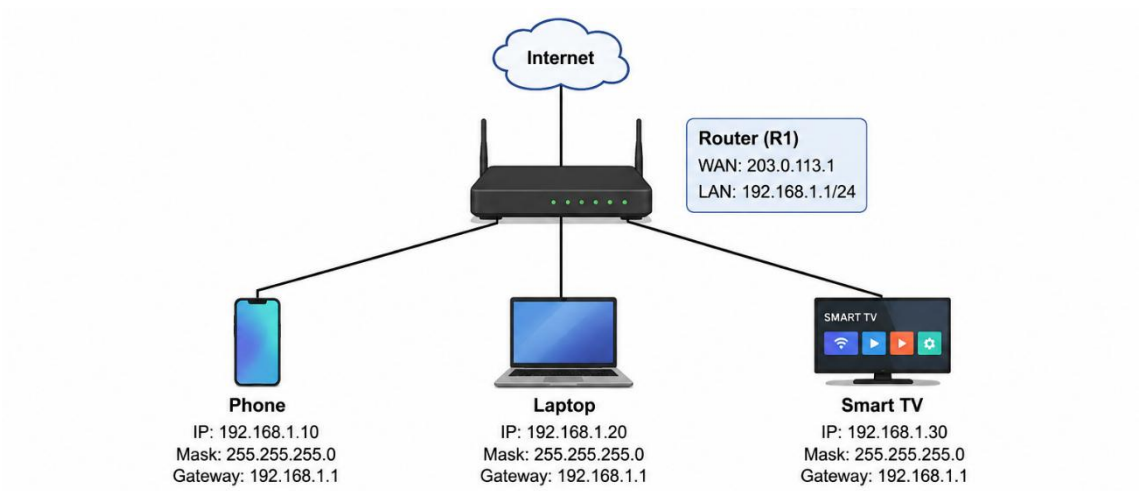


IP routing and routing protocols

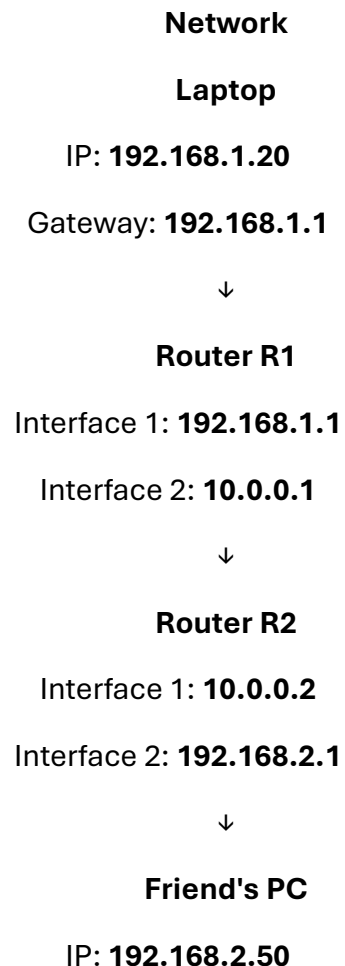
- 1) Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



Device	IP Address	Subnet Mask	Default Gateway
Router LAN	192.168.1.1	255.255.255.0	-
Router WAN	203.0.113.1	Public IP	ISP
Phone	192.168.1.10	255.255.255.0	192.168.1.1
Laptop	192.168.1.20	255.255.255.0	192.168.1.1
Smart TV	192.168.1.30	255.255.255.0	192.168.1.1

- 2) Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.



Packet Journey

Step 1

Laptop wants to send data to **192.168.2.50**.

Step 2

Laptop checks:

Destination = **192.168.2.50**

My subnet = **192.168.1.0/24**

Different subnet → Send packet to default gateway **192.168.1.1**

Step 3

Router R1 receives the packet.

Routing Table:

Destination	Next Hop
192.168.2.0/24	10.0.0.2

R1 forwards the packet to Router R2.

Step 4

Router R2 receives the packet.

Routing Table:

Destination	Interface
192.168.2.0/24	LAN

R2 sends the packet directly to **192.168.2.50**.

Router R1

Destination	Subnet Mask	Next Hop	Interface
192.168.1.0	255.255.255.0	Direct	LAN
10.0.0.0	255.255.255.252	Direct	WAN
192.168.2.0	255.255.255.0	10.0.0.2	WAN

Router R2

Destination	Subnet Mask	Next Hop	Interface
192.168.2.0	255.255.255.0	Direct	LAN
10.0.0.0	255.255.255.252	Direct	WAN
192.168.1.0	255.255.255.0	10.0.0.1	WAN

3) Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Branches

- Ahmedabad
- Mumbai
- Delhi

Each branch uses a different subnet.

Network Details

Branch	Network
Ahmedabad	192.168.10.0/24
Mumbai	192.168.20.0/24
Delhi	192.168.30.0/24

Routing Table

Network Address	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	Direct	Gig0/0
192.168.20.0	255.255.255.0	10.1.1.2	Gig0/1
192.168.30.0	255.255.255.0	10.1.1.6	Gig0/2

- 4) Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Static Routing	Dynamic Routing (RIP/OSPF)
Configured manually.	Routes are learned automatically.
Best for small networks.	Best for medium and large networks.
Does not adapt automatically.	Automatically updates routes if links fail.

Static Routing

Pros

1. Easy to understand.
2. Uses less CPU and memory.
3. More secure because routes are manually configured.

Cons

1. Time-consuming for large networks.
2. Doesn't automatically recover from failures.
3. Difficult to maintain.

Example: A small office with only one router.

Dynamic Routing (RIP/OSPF)

Pros

1. Automatically discovers routes.
2. Quickly adapts to network failures.
3. Easy to manage in large networks.

Cons

1. Uses more CPU and bandwidth.
2. More complex to configure.
3. Takes time to converge after changes.

Example: Food delivery companies like Zomato use dynamic routing in large networks so data can continue flowing even if one connection fails.

- 5) Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

Router R1

enable

configure terminal

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

write memory

Router R2

enable

configure terminal

router ospf 1

network 192.168.2.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

write memory

Configuration Explanation

Command	Description
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
router ospf 1	Starts the OSPF process with process ID 1.
network 192.168.1.0 0.0.0.255 area 0	Advertises the 192.168.1.0/24 network in OSPF Area 0.
network 10.0.0.0 0.0.0.3 area 0	Advertises the link between the two routers.
end	Exits configuration mode.
write memory	Saves the configuration so it remains after reboot.